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Advanced Risk Assessment and Mitigation Techniques for Semiconductor Manufacturing and Supply Chain Resilience

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Abstract
Semiconductor fabrication includes intricate processes susceptible to errors; conventional Fault detection and diagnosis (FDD) techniques find it difficult to handle nonlinear, dynamic, and uncertain process dynamics. To address this issue, this study proposes a Probabilistic Empowering Spiking Decision Neural Networks with Pine Cone Optimization Algorithm (PRSDNNets+PCOA) approach to FDD in semiconductor production. The input data are collected from the SECOM dataset and these data are preprocessed by the Adaptive and Propagated Mesh Filtering (APMF) technique. After preprocessing, the feature extraction process is performed by the A New Discrete Cosine-Krawtchouk-Tchebichef Transform (ANDCKTT). Fault detection and diagnosis using Probabilistic Empowering Spiking Decision Neural Networks (PRSDNNets) and optimization through Pine Cone Optimization Algorithm (PCOA). The efficiency of the proposed PRSDNNets+PCOA model is tested with an accuracy rate of 99.9% and specificity of 99.8%. The proposed model is realized using the Python programming language. The result of the proposed approach provides credible fault detection, improved diagnostic accuracy, and efficient treatment of nonlinear and uncertain manufacturing process behaviors.
Keywords: *Adaptive and Propagated Mesh Filtering, A New Discrete Cosine-Krawtchouk-Tchebichef Transform, Fault detection and diagnosis, Probabilistic Empowering Spiking Decision Neural Networks, Pine Cone Optimization Algorithm.*

1. Introduction

The semiconductor manufacturing process requires fault detection and diagnosis (FDD) methods to ensure reliable functioning of production operations [1]. Modern semiconductor devices become more complicated which makes minimal manufacturing process flaws lead to significant performance problems or yield decreases. High-quality standards need effective FDD techniques which allow early detection of defects and their diagnoses thus enabling prompt corrective measures [2]. The semiconductor fabrication techniques including photolithography and etching together with deposition and doping demand high precision because they easily respond to changes in process conditions. Sophisticated systems that perform real-time monitoring and analysis are necessary for detecting faults in the production process since these steps occur [3]. Sophisticated data analytics combined with machine learning technology and sensor systems assist FDD systems in detecting anomalies and fault classification as well as failure prediction prior to production downtime [4]. Machine manufacturers depend on FDD systems because they contribute to minimizing downtime and reducing defects and improving overall yield performance in the competitive semiconductor industry. Manufacturers who deploy efficient fault detection and diagnosis methods achieve improved process optimization while ensuring high manufacturing flow [5].

Existing challenges and methods include high sensor data limitations, complicated fault patterns, and high costs of real-time monitoring systems. The current work tries to address the aforementioned issues via its motivation.

Novelty and Contribution

Below are the novelty and contributions of the paper:

- To develop an advanced fault detection system using PRSDNNets+PCOA in semiconductor manufacturing processes.
- To preprocess SECOM dataset using APMF for improved fault detection in semiconductor manufacturing.
- To extract features from preprocessed data using ANDCKTT for improved fault detection in the production of semiconductors.
- To enhance FDD in semiconductor manufacturing by optimizing PRSDNNets using the PCOA.

The remaining manuscript is divided into 5 sections: Section Two, literature review; Section Three, suggested methods; Section Four, results and discussion; and Section Five, conclusion and upcoming projects.

2. Related Works

The papers related to neural network methods for FDD in Semiconductor Manufacturing are given below:

In 2019, Zhang, et al [6] has introduced a multiple time-series convolutional neural network (MTS-CNN) framework is suggested in the study for semiconductor manufacturing problem diagnosis and detection. Sliding windows are used for data augmentation in order to prevent over-fitting and increase diversity. Through a diagnostic layer, the model learns the characteristics of the equipment sensors and determines the significance of each sensor. Its effectiveness in wafer production and fault identification is validated by the investigation.

In 2020, He, R.,, et al [7] has introduced a long short-term memory with attention mechanism (LSTM-AM) model is used in the research to provide a low-yield underlying cause search approach for classifying semiconductor product data. Based on weights of attention taken from actual yield data from an electronic semiconductor business, experiments demonstrate that the strategy performs better than current approaches in forecasting low yield.

In 2019, Shen, Y, et al [8] has introduced a Convolutional Transformers and 1D Residual Networks (CT-1D ResNet) are compared in this study for fault identification with raw sensor data. The 1D ResNet operates well in classification tasks, whereas the Convolutional Transformers capture temporal and spatial connections. Research conducted on operational semiconductor industry data points to successful fault detection results through these analysis methods. The evaluation reveals how training parameters in addition to input representation, network architecture and their impact on performance.

Problem statement

In semiconductor production, timely and precise FDD are important to guarantee product quality, reduce downtime, and lower production costs. Conventional approaches tend to falter in dealing with intricate high-dimensional data and dynamic production conditions. The current work introduces a new method employing Probabilistic Empowering Spiking Decision Neural Networks with Pine Cone Optimization Algorithm (PRSDNNets+PCOA). This hybrid framework is designed to enhance detection accuracy, response time, and reliability in highly automated semiconductor production processes.

3. Proposed Methodology

This paper presents the Probabilistic Empowering Spiking Decision Neural Networks with Pine Cone Optimization Algorithm (PRSDNNets+PCOA), a new and efficient technique for FDD in semiconductor production. The proposed method adheres to a systematic five-stage framework that is intended to enable thorough analysis: (1) Data Collection, (2) Preprocessing, (3) Feature Extraction, (4) Fault detection and diagnosis, and (5) Optimization. The working architecture of PRSDNNets+PCOA is shown in Figure 1.

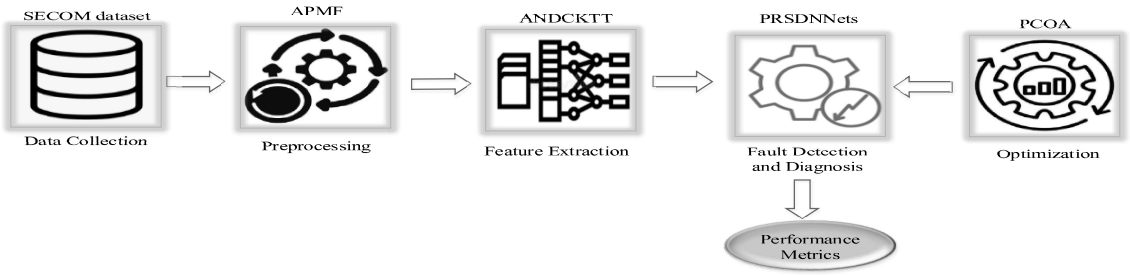

Figure 1: The operational architecture of PRSDNNets+PCOA

3.1 Data Collection

In the research the input data are gathered from the SECOM dataset. The SECOM (Semiconductor Manufacturing) dataset is utilized in order to explore the production process and evaluate the semiconductor quality. It contains data gathered from a wafer fabrication manufacturing line, having 1,567 instances. An instance contains 590 manufacturing operation variables sensed by sensors and a single quality output as either pass or fail. The quality label is binary: a fail is represented by 1 (positive), and a pass is represented by -1 (negative). Out of the 1,567 samples, only 105 are labeled as failures, creating a significant class imbalance with a fail-to-pass ratio of approximately 1:14. In addition to this imbalance, the dataset contains some missing values. The main objective of working with the SECOM dataset is to predict whether a semiconductor produced through this manufacturing process will be defective or not.

3.2. Adaptive and Propagated Mesh Filtering (APMF) Based Pre-processing

Preprocessing is crucial for enhancing the accuracy of FDD tasks in semiconductor manufacturing by refining complex, noisy sensor data. The Adaptive and Propagated Mesh Filtering (APMF) [9] technique plays a key role, improving data quality through noise filtering and vertex position updating. By treating the feature-space as a mesh, APMF refines structural representations, enhancing fault pattern identification in datasets like SECOM. The face normal $\hat{o}_b$ of a mesh face is refined using the normals of its neighboring faces, as shown in equation (1):

$$\hat{o}_b = \frac{1}{A_b} \sum_{s_b \in X_1} I_\sigma (Q(c_b, c_g)) I_{v_b} (Q(o_b, o_g)) o_g, \forall s_b \in T \tag{1}$$

where, $\hat{o}_b$ denotes the neighborhood of face s_b, $Q(c_b, c_g)$ represents the geodesic distance between centroids, and $Q(o_b, o_g)$ is the normal gradient distance. Gaussian weights I_σ and I_{v_b} are applied to preserve feature continuity while suppressing noise. A globally adaptive parameter σ is introduced as the average distance between centroids given by equation (2):

$$\sigma = j_e \tag{2}$$

To further refine this adaptiveness, σ is calculated based on entropy and the normal difference between neighboring elements. APMF denoises and smooths SECOM data, enabling accurate, adaptive fault detection without manual tuning across varying conditions. The preprocessed input data are provided for feature extraction according to the details described below,

3.3 A New Discrete Cosine-Krawtchouk-Tchebichef Transform (ANDCKTT) based Feature Extraction

Following the completion of data preprocessing, the feature extraction process is carried out using the ANDCKTT [10] algorithm. This hybrid method, ANDCKTT, combines the Discrete Cosine Transform (DCT), Krawtchouk Transform, and Tchebichef Transform to efficiently represent multimodal sensor data in semiconductor manufacturing. The ANDCKTT basis matrix is constructed by combining the individual transform matrices using mathematical formulations equations (3 and 4):

$$W = D_x D_y D_z D_y D_z \tag{3}$$

$$W = D_x (D_y D_z)^2 = D_{dxyzyz} \tag{4}$$

where, D_x the DCT basis components, D_y Krawtchouk, and D_z Tchebichef matrices are synthesized to form a robust hybrid transform. The ANDCKTT transform efficiently compresses data by focusing on high-energy coefficients, preserving key fault features. The extracted features are used to FDD, as defined in the subsequent sections.

3.4 Probabilistic Empowering Spiking Decision Neural Networks (PRSDNNets) Based Fault Detection and Diagnosis

After feature extraction, the FDD is carried out. It contains a Probabilistic Spiking Neural Networks (PSNN) with Empowering Decision Transformer (EDT). In FDD in Semiconductor Manufacturing, we introduce a Probabilistic Empowering Spiking Decision Neural Networks (PRSDNNets) framework. The follow-up is described in detail in the next section.

3.4.1 Probabilistic Spiking Neural Networks (PSNN)

PSNNs [11] leverage biologically inspired spiking neurons with probabilistic firing mechanisms to model complex, time-dependent processes. In semiconductor manufacturing, PSNNs are particularly

effective for FDD, as they can handle multivariate time-series data originating from sensors and control systems across fabrication equipment.

- **Spiking Neuron Model**

The PSNN framework typically uses the Leaky Integrate-and-Fire (LIF) neuron model, where the membrane potential evolves according to the dynamics defined by equation (5):

$$h_c^t(t) = \left(1 - \frac{1}{\mu}\right) a_c^t(t-1) + L_c^t(t) \tag{5}$$

where, $h_c^t(t)$ the membrane potential before firing, $a_c^t(t-1)$ depends on the neuron's previous membrane potential, a_c^t a bias term, and μ the membrane time constant, L_c^t is a weight parameter. When the potential crosses a firing threshold, the neuron fires probabilistically based on a Boltzmann distribution described in equations (6) and (7):

$$Q(e_c^t(t)) = \frac{d^{-W(e_c^t(t))}}{\sum_{e_c^t(t)\in\{0,1\}} d^{-W(e_c^t(t))}} \tag{6}$$

$$Q(e_c^t(t)=1) = 1 - \varsigma(G_{th} - h_c^t(t)) \tag{7}$$

where $\varsigma(z) = \frac{1}{1+e^{-z}}$ the firing probability is determined using a sigmoid activation function. This probabilistic behavior captures uncertainty and variability in process signals, which is especially relevant in high-precision manufacturing environments. In this PSNN-based FDD framework, data from manufacturing tools is analyzed using ADM to highlight key fault indicators like sudden deviations or pattern shifts. ADM enhances sensor clarity and detection accuracy, making PSNNs effective for detecting faults in real time and predictive maintenance in semiconductor manufacturing.

3.4.2 Empowering Decision Transformer (EDT)

The EDT [12] adapts the core autoregressive trajectory modeling of the Decision Transformer (DT) for FDD in semiconductor manufacturing. Unlike DT, which focuses on reinforcement learning, EDT predicts fault types or severity levels using historical multimodal data (sensor readings, equipment logs, and operational parameters). In standard DT, trajectories follow the form given by equation (8):

$$\mu = \left(\hat{A}_1, s_1, g_1, \hat{A}_2, s_2, g_2, \ldots, \hat{A}_N, s_N, g_N\right) \tag{8}$$

Each modality (e.g., temperature, pressure, acoustic signals) is encoded through dedicated encoders and projected into a shared token space, processed by a GPT-style transformer. To improve detection and diagnosis accuracy while reducing error rates, processing time, complexity, and cost, PCOA is applied to optimize the PRSDNNets model for FDD in Semiconductor Manufacturing. The optimization process is represented by the PRSDNNets model as follows:

3.5 Optimization using the Pine Cone Optimization Algorithm (PCOA)

The Pine Cone Optimization Algorithm (PCOA) [13] improves the performance of the PRSDNNets model by optimizing its hyperparameters, including biases, to enhance accuracy while reducing costs, processing time, and computational complexity. PCOA is a population-based, nature-inspired optimization method that mimics processes like pollination, gravity, and animal seed dispersal to effectively explore and exploit search spaces. A detailed illustration of the PCOA procedure is provided in Figure 2.

Step1: Initialization

The initial population of pine trees and cones is generated, with cones randomly placed around trees. It is given by equation (9):

$$GX_{e,c} = IgC_c + rn\vec{d}_{1\times\dim} \times (rnd \times upper\ boundsC_c - rnd \times Lower\ boundC_c) \tag{9}$$

Step 2: Generation of Random Variables

Random variables are generated in this step to simulate natural randomness.

Step 3: Evaluation of Fitness Function

The fitness function is specifically designed to improve model performance by minimizing the error rate and maximizing accuracy. The fitness function is expressed by equation (10):

$$f = \min imize\left(a_c^t\right)\max imize\left(L_c^t\right) \tag{10}$$

where, a_c^t the objective is to reduce costs, computational overhead, and error rates while simultaneously L_c^t improving accuracy.

Step 4: Exploration (Pollination Phase)

In the exploration phase, pollen particles mimic wind pollination, and cones move between trees to explore new solutions. It is given by equation (11):

$$GX_{e,c}^{nw} = GX_{e,c} + 0.5 \times v_{r_1} \times (Ibest_e - GX_{r_1,}) + 0.5 \times v_{r_3} \times (Ibest_e - GX_{r_3,c}) \tag{11}$$

Step 5: Exploitation (Cone Dispersal Phase)

Exploitation is the local search phase, where mature cones are dispersed near parent trees, refining solutions through gravity-based updates. It is given by equation (12):

$$GX_{e,c}^{nw} = \begin{cases} IW_e + v_1 \times R_1 \times (R_2 \times (upper\, c_r^e - Lowerc_r^e - IW_e)) & if\ control_{pramter} = 0 \\ GX_{e,c} + v_1 \times R_1 \times (R_2 \times (upper\, c_r^e - Lowerc_r^e - I_{pop_{al,s_1}}) - Ibest_{e,c}, otherwise) \end{cases} \tag{12}$$

where, v_1 is an adaptive weight, R_1 *and* R_2 represents random variables, and $I_{pop_{al,s_1}}$ is the random memory solution representing the best solution in the segment.

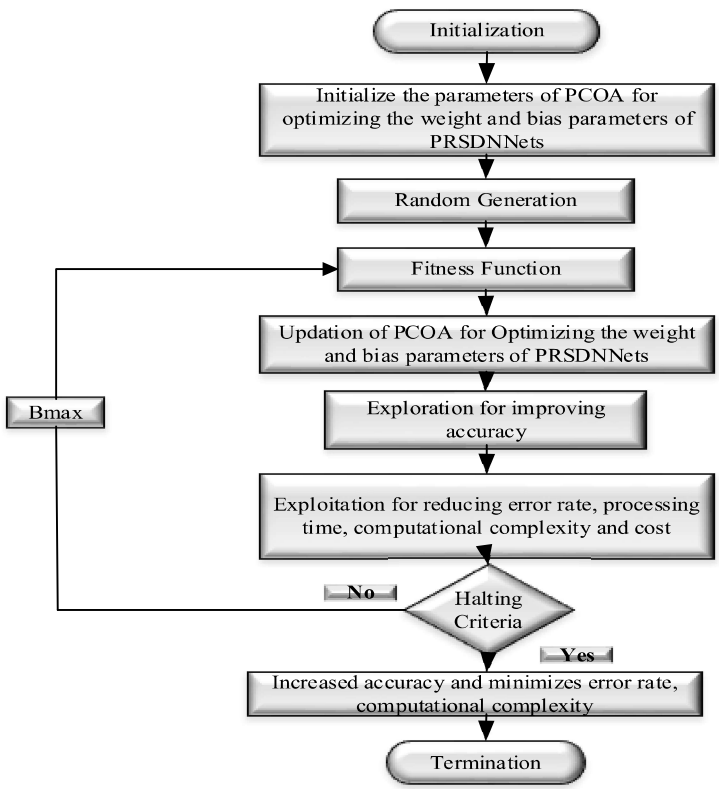

Figure 2: A detailed illustration of the PCOA procedure

- **Animal-Based Operators:**

A key part of exploitation is the animal-based operators, such as squirrel burial, which utilizes quadratic programming. This is given by equation (13):

$$W_{ini} = W_{best} + rnd \times (\overline{G}\,\overline{W} - W_{best}) \tag{13}$$

Step 6: Termination

The PCOA process continues iteratively until a specified maximum generation $B\max$ is reached or until fitness convergence occurs. The proposed Probabilistic Empowering Spiking Decision Neural Networks with Pine Cone Optimization Algorithm (PRSDNNets+PCOA) is designed for FDD in Semiconductor Manufacturing. The process begins with data preprocessing using APMF techniques, followed by feature extraction through ANDCKTT. The PRSDNNets model provides highly accurate fault detection and diagnosis, with its performance further enhanced through PCOA-based optimization. These results were then sequentially presented and discussed in the subsequent sections.

4. Result and Discussions

This section presents the results obtained by applying the proposed methodology using the Python platform. Key implementation parameters and experimental conditions are summarized in Table 1.

Table 1: Implementation Parameters

Parameters	Values
Proposed Neural Network	PRSDNNets
OS	Windows 10
Optimization	PCOA
Dataset	SECOM
Software	Python 3.7

Figure 3 shows the (a) Training vs. testing error and (b) loss comparison across multiple models.

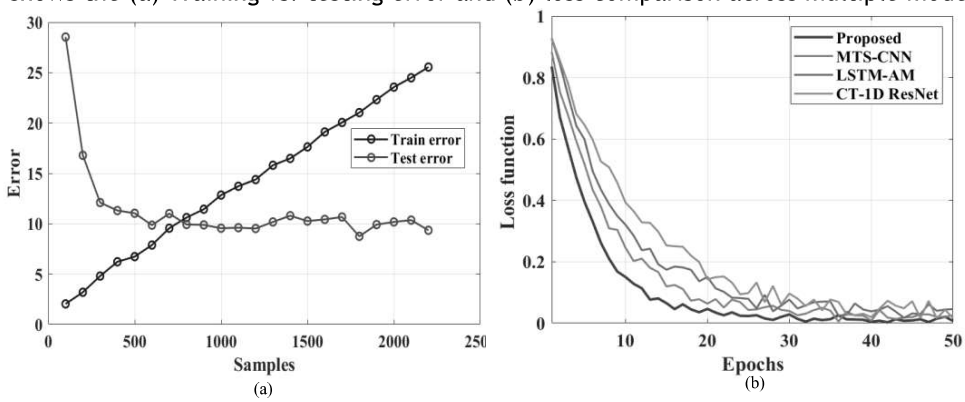

Figure 3: The (a) Training vs. testing error and (b) loss comparison across multiple models

Figure 3 illustrates two performance metrics: (a) shows training vs. testing error, where training error increases with more samples, indicating potential overfitting; (b) presents loss function comparison across models, where the proposed method outperforms others (MTS-CNN, LSTM-AM, CT-1D ResNet) by achieving faster convergence and lower final loss values. A thorough assessment of the model's efficiency utilizing the used techniques is shown in Table 2.

Table 2: Comparison of the Model's Performance against Existing Techniques

Methods / **Performance**	Accuracy (%)	Specificity (%)	Precision (%)	Computational time (sec)	Error rates (%)	F1-score (%)
MTS-CNN [6]	97.9	90.7	93.6	0.80	9.3	90.2
CuSO-MuLSTM [7]	96.0	93.9	95.3	0.91	7.0	95.3
WBANs [8]	92.6	96.0	96.3	0.86	9.2	94.3
PRSDNNets+PCOA (Proposed)	**99.9**	**99.8**	**99.8**	**0.20**	**0.1**	**99.8**

Table 2 compares the performance of various models, highlighting that the proposed PRSDNNet+PCOA significantly outperforms others. It achieves the highest accuracy (99.9%), specificity (99.8%), precision (99.8%), and F1-score (99.8%), with the lowest error rate (0.1%) and fastest computation time (0.20 seconds), demonstrating superior efficiency and accuracy.

5. Conclusion

In this manuscript, the PRSDNNet+PCOA is successfully implemented. The input data from the SECOM dataset and are preprocessed with the APMF method and then further feature extraction with the ANDCKTT method. Fault detection and diagnosis is performed through the PRSDNNet model, and then optimized with the PCOA to improve performance. The system is coded in Python. The proposed PRSDNNet+PCOA model provides excellent performance, with 99.9% accuracy and an error rate of as low as 0.1%. Future work could focus on optimizing algorithm efficiency, real-time implementation, and expanding fault detection to diverse semiconductor processes.

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